PROJECT REQUIREMENTS / VERSION 1.0 / 23 OCTOBER 2026

Decathlon Tableau Analytics

Combined DAD, BRD & FRD

Dataset Assessment Document • Business Requirements Document • Functional Requirements Document

A requirements baseline derived from the supplied **v12.twbx** workbook and the Tableau Intern Project Sprint brief. It describes the existing design and the work needed to verify and complete it.

Document control	Value
Project	Decathlon retail, inventory and manufacturing analytics
Repository	A-Rafeef/decathlon-Tableau-Project-
Status	Draft for stakeholder review; requirements and targets are proposed unless identified as observed.
Evidence	Static ZIP and XML inspection of v12.twbx; sprint requirements from the supplied DOCX.
Approval	Business owner, data owner and QA reviewer: to be assigned. No approval is claimed.

Observed workbook baseline

Four dashboards, 27 worksheets, ten logical tables, nine relationships and 21 calculated fields. The package contains one Hyper extract and four image assets. Its ZIP integrity check passed and workbook XML parsed successfully. The recorded authoring build is Tableau 2026.2.3 on Windows.

What this document establishes

The workbook supports four areas: executive sales performance, product and customer analysis, inventory risk, and manufacturing quality. This document maps those areas to business questions, data requirements, functional behavior and measurable acceptance checks.

Evidence boundary

This is a retrospective specification, not evidence that a Day-1 document or stakeholder sign-off previously existed. Dataset rows were not queried. Record counts, date coverage, duplicates, missingness, KPI values, refresh behavior and Tableau interactions remain unverified. Configured features are not treated as passed runtime tests.

Reading guide: DAD on pages 2–3; BRD on page 4; KPI requirements on pages 5–6; FRD on pages 7–8; acceptance and source register on page 9.

Source inventory & intended grain

The federated source is named **dim_products+**. Its model references six dimension tables and four fact tables. These are logical relationships, not evidence of one flattened, joined dataset. Candidate grains below are inferred from field names and require uniqueness checks.

Logical table	Candidate grain / key	Relevant fields
dim_products.csv	One product / product_id	product_name, sport_department, passion_brand, unit_cost, unit_price
dim_customers.csv	One customer / customer_id	loyalty_tier, primary_sport_interest, signup_date, customer_name
dim_date.csv	One calendar date / date	year, month, quarter, is_weekend, holiday_season
dim_stores.csv	One store / store_id	store_name, store_format, floor_size_sqm, opened_date
dim_suppliers_plants.csv	One facility / facility_id	facility_name, facility_type, max_monthly_capacity_units
dim_workers.csv	One worker / worker_id	store_id, role, monthly_salary, performance_rating
fact_sales_transactions.csv	Transaction or order line / transaction_id; order_id may repeat	order_date, product_id, store_id, customer_id, quantity, net_revenue, gross_profit
fact_inventory.csv	Store-product-date snapshot / snapshot_id	snapshot_date, stock_on_hand, safety_stock_threshold, stockout_flag
fact_production.csv	Production batch / batch_id	production_date, facility_id, product_id, actual_quantity, planned_quantity, defect_quantity
fact_financial_pl.csv	Store-month P&L / financial_record_id	fiscal_month, store_id, net_revenue, operating_profit, net_operating_profit

Package and refresh constraints

The archive includes **v12.twb**, a 19,922,944-byte Hyper extract at **Data/v1.twb Files/dim_products+.hyper**, and design images. Original CSV files are not packaged. CSV connections refer to an author-local Windows directory, so refreshing from CSV requires the source files and relocated connections. Opening the embedded extract still requires a Tableau runtime check.

Dataset row counts, date ranges, refresh cadence, source owner, licensing and real-versus-synthetic provenance are not established by the inspected metadata.

Relationships, quality & controls

Observed relationship	Equality field
Sales → Customers	customer_id
Sales → Date	order_date = date
Sales → Stores	store_id
Production → Facilities	facility_id
Stores → Workers	store_id
Stores → Financial P&L	store_id
Stores → Inventory	store_id
Products → Production	product_id
Products → Sales	product_id

Model implications

The date dimension relates directly only to sales. Production dates, inventory snapshots and financial months need explicit filter semantics. Inventory has product_id but no direct relationship to Products. Sales and financial P&L share Stores without a fiscal-month relationship. Test cross-fact totals and filters at their own grains before making comparisons.

Required data assessment before acceptance

Check	Proposed acceptance rule
Keys and grain	No unexplained null or duplicate primary keys. Verify store-product-date snapshots and store-month P&L uniqueness.
Relationships	Count unmatched keys for every relationship; resolve or explicitly document exclusions. Validate declared store uniqueness against rows.
Completeness and time	Profile nulls and date coverage per table. Confirm refresh date, reporting period, missing months and snapshot cadence.
Numeric validity	Reconcile source and dashboard totals; investigate negative amounts, returns, discounts, invalid capacities and zero denominators.
Currency and units	Establish a single reporting currency or approved conversion. “Cr” means division by 10,000,000; it does not establish INR.

Data handling

Schema includes customer names, worker names, salary and performance fields. Confirm provenance and sharing rights before publishing records or an embedded extract. This document contains schema and requirements only. Customer and workforce row-level reporting is outside the proposed dashboard scope.

Business purpose & success criteria

Problem and objective — proposed

Provide a consistent view of sales, profitability, inventory risk and production performance so business users can identify areas needing investigation. Reduce ambiguity caused by mixed table grains, reporting periods and KPI definitions. No realized business benefit or financial outcome is claimed.

ID / stakeholder	Business question and intended decision	Evidence / delivery
BR-01 Executive / sales lead	How do net sales, profit, orders and AOV vary by time, store, channel and department? Prioritize performance reviews.	Overview dashboard; revenue and order trends; channel and department views.
BR-02 Merchandising / customer analyst	Which products combine revenue, margin and discount exposure? How does revenue vary by sport interest?	Product & Customer Insights; product profitability and customer-interest views.
BR-03 Inventory / store manager	Where is stock below safety levels, and how do stock trends vary by store? Prioritize replenishment review.	Inventory Risk Detail; snapshot summaries and monthly stock trend.
BR-04 Manufacturing / quality lead	How do actual output, planned output, capacity and defects vary by facility and period?	Manufacturing & Quality Analytics; attainment, utilization and defect metrics.
BR-05 Finance / project reviewer	Are definitions, periods, currency, controls and limitations explicit and reproducible?	Combined requirements, KPI register, reconciliation and QA record.

Scope and boundaries

In scope: the four existing analytical areas, approved filters and navigation, documented calculations, data checks, screenshots and final packaged workbook. **Outside this baseline:** transactional write-back, live enterprise deployment, forecasting claims, workforce evaluation and customer-level targeting. Forecasting or clustering requires a separate supported business question and dataset assessment.

Measurable completion criteria — proposed

All five business requirements map to a dashboard or validation artifact. Counts reconcile exactly; currency totals reconcile to the declared display precision and percentages to 0.01 percentage points for identical scope. All required interaction scenarios pass. No unresolved critical data or calculation issue remains. Three to five independently supported insights and the sprint Story are delivered before final submission.

Dependencies and decisions

Assign stakeholder owners; confirm source permissions, currency, period coverage, customer repeat-purchase definition, capacity denominator and latest-snapshot policy. These decisions are pending. Numeric business targets and operational SLAs have not been supplied.

Sales & customer KPI register

The formulas below reflect definitions observed in workbook XML. Internal calculation identifiers are replaced with their display names for readability. Their presence does not establish numeric correctness. Percentages are ratios displayed as percentages; all metrics require a documented date and filter scope.

Metric	Observed definition	Business rule / validation requirement
Net Sales (Cr)	SUM(net_revenue) / 10,000,000	Sales fact. Confirm reporting currency and sign treatment for returns.
Gross Profit (Cr)	SUM(gross_profit) / 10,000,000	Sales fact. Reconcile profit and cost definitions to the source.
Gross Margin % / Product Gross Margin	SUM(gross_profit) / SUM(net_revenue)	Weighted ratio of totals, not average of row margins. Define zero-revenue display.
Total Orders	COUNTD(order_id)	Count distinct orders rather than transaction lines. Validate null IDs and scope.
AOV	SUM(net_revenue) / Total Orders	Currency per distinct order. Reconcile order count and net revenue under the same filters.
Discount Rate / Product Discount Rate %	SUM(discount_amount) / SUM(gross_revenue)	Validate discount signs and zero gross-revenue handling.
Profit per Unit	SUM(gross_profit) / SUM(quantity)	Currency per unit. Clarify net versus gross quantities and returned units.
Orders per Customer (LOD)	{ FIXED [customer_id] : COUNTD([order_id]) }	Define lifetime versus selected-period behavior. Ordinary dimension filters may not change FIXED results.
Repeat Purchase Rate	COUNTD(IF Orders per Customer > 1 THEN customer_id END) / COUNTD(customer_id)	Customer share with more than one order. Ensure numerator and denominator use the agreed population and period.
Annualized Revenue per Sqm	SUM(net_revenue) / 2 / MIN(floor_size_sqm)	Hard-coded two-year assumption. Validate period coverage and area aggregation; do not claim general date-filter compatibility.

Shared calculation acceptance

Use independently computed totals for the same population, date interval, currency and granularity. Cover unfiltered, single-store, single-month, multi-month, empty-result and zero-denominator selections. Record expected and actual values; do not treat a visible KPI as validated.

The workbook also contains financial P&L measures with similar names to sales measures. Keep their source-table identity explicit and label full-period financial views when they are not controlled by sales date selections.

Operations metrics & advanced logic

Metric / logic	Observed definition	Requirement / caveat
Production Attainment	SUM(actual_quantity) / SUM(planned_quantity)	Actual output as a share of plan. Define behavior when planned quantity is zero.
Weighted Defect Rate	SUM(defect_quantity) / SUM(actual_quantity)	Weight by actual output. Confirm units and whether defects can exceed output.
Energy per Actual Unit	SUM(energy_consumption_kwh) / SUM(actual_quantity)	kWh per unit. Validate zero output and the production period used.
Capacity Utilization	SUM(actual_quantity) / MIN(max_monthly_capacity_units)	Review denominator at facility-month, multi-month and multi-facility scopes. A single minimum capacity is not period-adjusted.
Stock Below Safety	IF stock_on_hand < safety_stock_threshold THEN 1 ELSE 0 END	Row-level flag. Define counted entities and null behavior before interpreting totals.
Latest snapshot flag	snapshot_date = { MAX(snapshot_date) }	Global latest date. Decide whether the requirement is latest globally, within selection, or per store-product.
Order Month	DATETRUNC('month', order_date)	Month bucketing for orders. Preserve year identity when comparing multiple years.
Running sum [Calculation1]	RUNNING_SUM([Net Sales (Cr)])	Validate addressing, partitioning, sort order and restart behavior before presenting a cumulative total.
Box	Empty string literal	Layout helper; exclude from business KPI definitions and totals.

Observed sprint coverage

FIXED LOD, table-calculation syntax, dashboard filters, navigation buttons and show/hide controls are present. Four dashboards have fixed 1300 × 800 layouts. Their rendering and runtime behavior were not tested in Tableau.

Sprint gaps to resolve

No parameter definitions, INCLUDE or EXCLUDE calculations, or Tableau Story were found. The sprint calls for these advanced calculation exercises and a Story with three to five evidence-based points. Add meaningful implementations or document an agreed exception. Do not invent business insights to fill the Story.

Conditional features

URL actions, set actions, forecasting, clustering and other optional techniques should be used only when they improve a defined question. Their absence is not automatically a defect. Parameter-driven analysis is a specific sprint requirement and remains a documented gap.

Dashboard requirements & traceability

Observed worksheet and dashboard names are retained to make the document traceable. The functional statements below define proposed acceptance behavior; no live interaction has been certified.

Requirement / exact dashboard	Required content and behavior	Trace / verification
FR-01 p1-overview	Display sales, profit, margin, order and AOV KPIs; monthly revenue/order trend; channel and department breakdowns; KPI sparklines. Provide store and order-month controls.	BR-01. Reconcile KPI values and selected scope. Financial chart “Operating Profit by Store – Full Period” must retain an explicit scope label.
FR-02 P2-PRODUCT & CUSTOMER INSIGHTS	Show product revenue versus margin, discount versus margin, gross-profit ranking and revenue by customer sport interest. Provide order-date and sport-department controls.	BR-02. Verify product ranking, metric axes and common filter coverage. Customer-interest view has no direct filter definitions observed; test propagation.
FR-03 p3-Inventory Risk Detail	Show store-level inventory summary, inventory risk, stock KPIs and monthly stock trend. Provide store and snapshot-date controls with an explicit latest-snapshot policy.	BR-03. Latest-date logic appears on risk, summary and KPI sheets, but not trend. Test earlier date ranges and missing latest snapshots.
FR-04 p4-Manufacturing & Quality Analytics	Show production attainment, planned versus actual output, facility defect rate, capacity utilization and energy intensity. Provide production-month controls.	BR-04. Reconcile facility-month and total-level results. Verify capacity and date aggregation behavior.
FR-05 Navigation and layout	Navigation buttons must route to the intended four pages. Show/hide controls must preserve access to filters. Labels and tooltips must identify units, scope and calculation meaning.	BR-01–04. Verify every button, back/next path, panel toggle and tooltip in Tableau at 1300 × 800.

Filter contract

Each control needs a documented list of target worksheets. Sales dates apply to order_date; inventory dates to snapshot_date; production dates to production_date. Do not imply that sales dates filter fiscal-month P&L. Cross-page filter persistence must be stated and tested rather than assumed.

Empty and exceptional states

Selections with no rows must show an understandable empty state without implying zero sales or zero defects. Zero denominators must use an agreed blank or “N/A” display. Titles, tooltips and totals must remain consistent with the active selection.

Interactions & nonfunctional criteria

Requirement	Observed evidence	Proposed acceptance behavior
FR-06 Channel selection	“1-Revenue by Channel” has an on-select filter action targeting p1-overview; auto-clear is true.	Selecting one or more channels updates the intended views; clearing restores the baseline. Confirm the financial view’s full-period scope.
FR-07 Department selection	“1-Revenue by Dept” has an on-select action targeting p1-overview; auto-clear is true.	Department selection and clearing update the intended worksheets without changing the meaning of totals.
FR-08 Inventory selection	“3-Store Inventory Summary” has an on-select action targeting p3-Inventory Risk Detail; auto-clear is true.	Store selection matches the store control, risk summary and trend. Clear the action and confirm baseline recovery.
FR-09 Residual actions	Worksheet filters reference Facility Name and Loyalty Tier actions whose IDs are absent from configured workbook actions.	Open in Tableau, inspect saved action filters and verify their effect. Remove or restore stale behavior if confirmed. Runtime impact is currently unknown.
FR-10 Story and parameter analysis	No Story or parameter definitions were found in the supplied XML.	Complete the sprint’s parameter-driven analysis and three-to-five-point Story using validated evidence; record an approved exception if scope changes.

Proposed nonfunctional requirements

ID / area	Acceptance condition
NFR-01 / Usability	No clipped chart titles, legends, tooltips or essential controls at the supported 1300 × 800 layout. Do not claim mobile support without testing it.
NFR-02 / Performance	Provisional targets: initial dashboard load ≤10 seconds; standard filter response ≤3 seconds on an agreed reference machine. Record cold/warm timings and get owner agreement.
NFR-03 / Portability	Final .twbx opens with its embedded extract on a compatible Tableau version. Document any refresh dependency on external CSV files.
NFR-04 / Reproducibility	Record workbook version, source refresh, metric definitions, independent check values, defects and fixes. Keep publication status truthful.
NFR-05 / Sharing	Publish only approved data and artifacts. Confirm names, salaries and embedded data are synthetic or authorized before distributing the workbook.

Validation plan & open decisions

Only archive integrity, XML parsing and structural inventory were executed for this document. All remaining cases below are pending. Expected results are proposed acceptance conditions, not recorded passes.

Test / trace	Scenario and expected result	Status
T-01 / DAD	ZIP CRC and workbook XML parse without errors; four dashboards and 27 worksheets inventoried.	Passed: static check
T-02 / DAD	Profile row counts, missingness, candidate keys and unmatched relationships; document every unresolved exception.	Not run
T-03 / BR-01–05	Recompute KPI totals independently for matching period, population, grain and currency; meet BRD reconciliation tolerances.	Not run
T-04 / FR-01–04	Exercise each visible filter, all/single/multiple selections and an empty result; only intended sheets change.	Not run
T-05 / FR-06–09	Select and clear every action; verify restored totals and investigate residual action references.	Not run
T-06 / FR-05, NFR-01	Check each navigation and show/hide button, tooltip, title and layout for correctness and clipping.	Not run
T-07 / KPI register	Test FIXED LOD filter order, running-sum partitions, annualization, monthly capacity and zero denominators.	Not run
T-08 / FR-03	Choose a period excluding the global latest snapshot; verify agreed snapshot behavior and trend consistency.	Not run
T-09 / NFR-02–03	Open packaged workbook on a compatible Tableau installation; record load/response timings and separately test CSV refresh.	Not run
T-10 / FR-10, NFR-05	Verify Story and parameter work, three to five evidenced insights and approved sharing/publication scope.	Pending delivery

Decisions required for sign-off

Business owner: approve KPIs, reporting currency, period scope and provisional targets. Data owner: confirm source provenance, sharing rights, refresh cadence and table grains. Analyst: resolve repeat-purchase and snapshot semantics, annualization and capacity logic. QA reviewer: record expected/actual results and close critical issues. Owners and approval dates remain unassigned.

Source register and evidence

S1: User-supplied v12.twbx; member v12.twb. Evidence paths: workbook/datasources/datasource, object-graph/relationships, column/calculation, worksheets, dashboards and actions. ZIP inspected without altering the source.
S2: UPDATED_TABLEAU_INTERN_PROJECT_SPRINT (1).docx; Final Submission Requirements, Eight-Day Work Plan, GitHub Repository Structure and Dashboard Acceptance Criteria. S2 defines desired outputs, not implementation evidence.

S1 SHA-256: 896b85f8ffabfd39c55636fa8760adc4d529bf1bc20e383e3ccf24a0cb2c1df1

Document scope: one combined requirements PDF. No raw records, source workbook, live Tableau Public link, screenshots or fabricated test results are included.